(**a**) Justify the equality

$$\frac{d}{dz}[f(z)g(z)] = \lim_{\Delta z \to 0} \frac{f(z+\Delta z)g(z+\Delta z) - f(z)g(z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left[\frac{f(z+\Delta z) - f(z)}{\Delta z} g(z+\Delta z) + f(z) \frac{g(z+\Delta z) - g(z)}{\Delta z} \right].$$

(**b**) Use Definition 3.1.2 to justify $\lim_{\Delta z \to 0} g(z+\Delta z) = g(z)$.

(**c**) Use Theorems 3.1.2(*ii*) and 3.1.2(*iii*) to finish the proof.

35. In this problem you will prove that $f(z) = \overline{z}$ is nowhere differentiable using the polar form $\Delta z = r(\cos\theta + i\sin\theta)$.

(**a**) What must r approach if $\Delta z \to 0$? Must θ approach a specific value?

(**b**) If $f(z) = \overline{z}$, show that

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{\cos\theta - i\sin\theta}{\cos\theta + i\sin\theta}.$$

(**c**) Explain succinctly why the result in part (b) shows that f is nowhere differentiable.

36. In this problem you will prove that $f(z) = |z|$ is nowhere differentiable using the alternative definition (12) and polar forms of z and z_0.

(**a**) Let $z_0 = r(\cos\theta + i\sin\theta)$ and $z = (r+\Delta r)(\cos\theta + i\sin\theta)$. Along this path determine $\lim_{z \to z_0} \dfrac{f(z) - f(z_0)}{z - z_0}$.

(**b**) Let $z_0 = r(\cos\theta + i\sin\theta)$ and $z = r\,[\cos(\theta + \Delta\theta) + i\sin(\theta + \Delta\theta)]$. Along this path determine $\lim_{z \to z_0} \dfrac{f(z) - f(z_0)}{z - z_0}$.

(**c**) Explain succinctly why the results in parts (a) and (b) show that f is nowhere differentiable.

37. Suppose $f'(z)$ exists at a point z. Is $f'(z)$ continuous at z?

38. (**a**) Let $f(z) = z^2$. Write down the real and imaginary parts of f and f'. What do you observe?

(**b**) Repeat part (a) for $f(z) = 3iz + 2$.

(**c**) Make a conjecture about the relationship between real and imaginary parts of f versus f'.

3.3 Cauchy-Riemann Equations

INTRODUCTION In the preceding section we saw that a function f of a complex variable z is *analytic at a point* z when f is differentiable at z and differentiable at every point in some neighborhood of z. This requirement is more stringent than simply *differentiability at a point* because a complex function can be differentiable at a point z but yet be differentiable nowhere else. A function f is *analytic in a domain* D if f is differentiable at all points in D. We shall now develop a test for analyticity of a complex function $f(z) = u(x, y) + iv(x, y)$ that is based on partial derivatives of its real and imaginary parts u and v.

A Necessary Condition for Analyticity

A review of partial derivatives can be found in Section 3.3 of the student study guide. ▶

In the next theorem we see that if a function $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point z, then the functions u and v must satisfy a pair of equations that relate their first-order partial derivatives.

THEOREM 3.3.1 Cauchy-Riemann Equations

Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z = x + iy$. Then at z the first-order partial derivatives of u and v exist and satisfy the **Cauchy-Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{1}$$

Proof The derivative of f at z is given by

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}. \tag{2}$$

By writing $f(z) = u(x, y) + iv(x, y)$ and $\Delta z = \Delta x + i\Delta y$, (2) becomes

$$f'(z) = \lim_{\Delta z \to 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \tag{3}$$

Since the limit (2) is assumed to exist, Δz can approach zero from any convenient direction. In particular, if we choose to let $\Delta z \to 0$ along a horizontal line, then $\Delta y = 0$ and $\Delta z = \Delta x$. We can then write (3) as

$$\begin{aligned} f'(z) &= \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y) + i\,[v(x + \Delta x, y) - v(x, y)]}{\Delta x} \\ &= \lim_{\Delta x \to 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}. \end{aligned} \tag{4}$$

The existence of $f'(z)$ implies that each limit in (4) exists. These limits are the definitions of the first-order partial derivatives with respect to x of u and v, respectively. Hence, we have shown two things: both $\partial u/\partial x$ and $\partial v/\partial x$ exist at the point z, and that the derivative of f is

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}. \tag{5}$$

We now let $\Delta z \to 0$ along a vertical line. With $\Delta x = 0$ and $\Delta z = i\Delta y$, (3) becomes

$$f'(z) = \lim_{\Delta y \to 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \to 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}. \tag{6}$$

In this case (6) shows us that $\partial u/\partial y$ and $\partial v/\partial y$ exist at z and that

$$f'(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \tag{7}$$

By equating the real and imaginary parts of (5) and (7) we obtain the pair of equations in (1). ☰

Because Theorem 3.3.1 states that the Cauchy-Riemann equations (1) hold at z as a *necessary* consequence of f being differentiable at z, we cannot use the theorem to help us determine where f is differentiable. *But* it is important to realize that Theorem 3.3.1 can tell us where a function f does not possess a derivative. If the equations in (1) are *not* satisfied at a point z, then f cannot be differentiable at z. We have already seen in Example 3 of Section 3.2 that $f(z) = x + 4iy$ is not differentiable at any

point z. If we identify $u = x$ and $v = 4y$, then $\partial u/\partial x = 1$, $\partial v/\partial y = 4$, $\partial u/\partial y = 0$, and $\partial v/\partial x = 0$. In view of

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = 4$$

the two equations in (1) cannot be simultaneously satisfied at any point z. Therefore, by Theorem 3.3.1, f is nowhere differentiable.

It also follows from Theorem 3.3.1 that if a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic throughout a domain D, then the real functions u and v satisfy the Cauchy-Riemann equations (1) at every point in D.

EXAMPLE 1 Verifying Theorem 3.3.1

The polynomial function $f(z) = z^2 + z$ is analytic for all z and can be written as $f(z) = x^2 - y^2 + x + i(2xy + y)$. Thus, $u(x, y) = x^2 - y^2 + x$ and $v(x, y) = 2xy + y$. For any point (x, y) in the complex plane we see that the Cauchy-Riemann equations are satisfied:

$$\frac{\partial u}{\partial x} = 2x + 1 = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

The contrapositive* form of the sentence preceding Example 1 is:

Criterion for Non-analyticity

If the Cauchy-Riemann equations are not satisfied at every point z in a domain D, then the function $f(z) = u(x, y) + iv(x, y)$ cannot be analytic in D.

EXAMPLE 2 Using the Cauchy-Riemann Equations

Show that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ is not analytic at any point.

Solution We identify $u(x, y) = 2x^2 + y$ and $v(x, y) = y^2 - x$. From

$$\begin{aligned} \frac{\partial u}{\partial x} &= 4x \quad \text{and} \quad \frac{\partial v}{\partial y} = 2y \\ \frac{\partial u}{\partial y} &= 1 \quad \text{and} \quad \frac{\partial v}{\partial x} = -1 \end{aligned} \tag{8}$$

we see that $\partial u/\partial y = -\partial v/\partial x$ but that the equality $\partial u/\partial x = \partial v/\partial y$ is satisfied only on the line $y = 2x$. However, for any point z on the line, there is no neighborhood or open disk about z in which the Cauchy-Riemann equations are satisfied. We conclude that f is nowhere analytic.

A Sufficient Condition for Analyticity By themselves, the Cauchy-Riemann equations do not ensure analyticity of a function $f(z) = u(x, y) + iv(x, y)$ at a point $z = x + iy$. It is possible for the Cauchy-Riemann equations to be satisfied at z and yet $f(z)$ may not be differentiable at z, or $f(z)$ may be differentiable at z but nowhere else. In either case, f is not analytic at z. See Problem 40 in Exercises 3.3. However, when we add the condition of continuity to u and v *and* to

*A proposition "If P, then Q" is logically equivalent to its contrapositive "If not Q, then not P."

the four partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$, it can be shown that the Cauchy-Riemann equations are not only necessary but also sufficient to guarantee analyticity of $f(z) = u(x, y) + iv(x, y)$ at z. The proof is long and complicated and so we state only the result.

THEOREM 3.3.2 Criterion for Analyticity

Suppose the real functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in a domain D. If u and v satisfy the Cauchy-Riemann equations (1) at all points of D, then the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in D.

EXAMPLE 3 Using Theorem 3.3.2

For the function $f(z) = \dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2}$, the real functions $u(x, y) = \dfrac{x}{x^2+y^2}$ and $v(x, y) = -\dfrac{y}{x^2+y^2}$ are continuous except at the point where $x^2 + y^2 = 0$, that is, at $z = 0$. Moreover, the four first-order partial derivatives

$$\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2}, \qquad \frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2},$$
$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}, \quad \text{and} \quad \frac{\partial v}{\partial y} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

are continuous except at $z = 0$. Finally, we see from

$$\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2} = -\frac{\partial v}{\partial x}$$

that the Cauchy-Riemann equations are satisfied except at $z = 0$. Thus we conclude from Theorem 3.3.2 that f is analytic in any domain D that does not contain the point $z = 0$. ☰

The results in (5) and (7) were obtained under the basic assumption that f was differentiable at the point z. In other words, if f is differentiable, then (5) and (7) give us a formula for computing the derivative $f'(z)$:

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}. \tag{9}$$

For example, we already know from part (i) of Theorem 3.2.1 that $f(z) = z^2$ is entire and so is differentiable for all z. With $u(x, y) = x^2 - y^2$, $\partial u/\partial x = 2x$, $v(x, y) = 2xy$, and $\partial v/\partial y = 2y$, we see from (9) that

$$f'(z) = 2x + i2y = 2(x + iy) = 2z.$$

For Example 3, the function $f(x) = \dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2}$ has derivative

$$f'(z) = \frac{y^2 - x^2}{(x^2+y^2)^2} + i\frac{2xy}{(x^2+y^2)^2}$$

in any domain that does not include $z = 0$.

Recall that analyticity implies differentiability but not conversely. Theorem 3.3.2 has an analogue that gives the following criterion for differentiability.

Sufficient Conditions for Differentiability

If the real functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in some neighborhood of a point z, and if u and v satisfy the Cauchy-Riemann equations (1) *at z, then the complex function $f(z) = u(x, y) + iv(x, y)$ is differentiable at z and $f'(z)$ is given by* (9).

EXAMPLE 4 A Function Differentiable on a Line

In Example 2 we saw that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ was nowhere analytic, but yet the Cauchy-Riemann equations were satisfied on the line $y = 2x$. Since the functions $u(x, y) = 2x^2 + y$, $\partial u/\partial x = 4x$, $\partial u/\partial y = 1$, $v(x, y) = y^2 - x$, $\partial v/\partial x = -1$, and $\partial v/\partial y = 2y$ are continuous at every point, it follows that f is differentiable on the line $y = 2x$. Moreover, from (9) we see that the derivative of f at points on this line is given by $f'(z) = 4x - i = 2y - i$. ☰

The following theorem is a direct consequence of the Cauchy-Riemann equations. Its proof is left as an exercise. See Problems 31 and 35 in Exercises 3.3.

THEOREM 3.3.3 Constant Functions

Suppose the function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D.

(*i*) If $|f(z)|$ is constant in D, then so is $f(z)$.

(*ii*) If $f'(z) = 0$ in D, then $f(z) = c$ in D, where c is a complex constant.

Polar Coordinates In Section 2.1 we saw that a complex function can be expressed in terms of polar coordinates. Indeed, the form $f(z) = u(r, \theta) + iv(r, \theta)$ is often more convenient to use. In polar coordinates the Cauchy-Riemann equations become

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}. \tag{10}$$

The polar version of (9) at a point z whose polar coordinates are (r, θ) is then

$$f'(z) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) = \frac{1}{r}e^{-i\theta}\left(\frac{\partial v}{\partial \theta} - i\frac{\partial u}{\partial \theta}\right). \tag{11}$$

See Problems 38 and 39 in Exercises 3.3.

Remarks COMPARISON WITH REAL ANALYSIS

In real calculus, one of the noteworthy properties of the exponential function $f(x) = e^x$ is that $f'(x) = e^x$. In (3) of Section 2.1 we gave the definition of the complex exponential $f(z) = e^z$. We are now in a position to show that $f(z) = e^z$ is differentiable everywhere and that this complex function shares the same derivative property as its real counterpart, that is, $f'(z) = f(z)$. See Problem 25 in Exercises 3.3.

EXERCISES 3.3

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1 and 2, the given function is analytic for all z. Show that the Cauchy-Riemann equations are satisfied at every point.

1. $f(z) = z^3$

2. $f(z) = 3z^2 + 5z - 6i$

In Problems 3–8, show that the given function is not analytic at any point.

3. $f(z) = \text{Re}(z)$

4. $f(z) = y + ix$

5. $f(z) = 4z - 6\bar{z} + 3$

6. $f(z) = \bar{z}^2$

7. $f(z) = x^2 + y^2$

8. $f(z) = \dfrac{x}{x^2 + y^2} + i\dfrac{y}{x^2 + y^2}$

In Problems 9–16, (**a**) use Theorem 3.3.2 to show that the given function is analytic in an appropriate domain, and (**b**) use (9) or (11) to find the derivative of the function in the domain.

9. $f(z) = e^{-x}\cos y - ie^{-x}\sin y$

10. $f(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$

11. $f(z) = e^{x^2-y^2}\cos 2xy + ie^{x^2-y^2}\sin 2xy$

12. $f(z) = 4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$

13. $f(z) = \dfrac{x-1}{(x-1)^2 + y^2} - i\dfrac{y}{(x-1)^2 + y^2}$

14. $f(z) = \dfrac{x^3 + xy^2 + x}{x^2 + y^2} + i\dfrac{x^2y + y^3 - y}{x^2 + y^2}$

15. $f(z) = \dfrac{\cos\theta}{r} - i\dfrac{\sin\theta}{r}$

16. $f(z) = 5r\cos\theta + r^4\cos 4\theta + i(5r\sin\theta + r^4\sin 4\theta)$

In Problems 17–20, find real constants a, b, c, and d so that the given function is entire.

17. $f(z) = 3x - y + 5 + i(ax + by - 3)$

18. $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$

19. $f(z) = 1 + 5r\cos\theta + ar\sin\theta + i\,(3 + br\sin\theta + r\cos\theta)$

20. $f(z) = r^a\cos 2\theta + br\sin\theta + i\left(cr^2\sin\theta\cos\theta + r\cos\theta\right)$

In Problems 21–24, (**a**) show that the given function is not analytic at any point but is differentiable along the indicated curve(s), and (**b**) use (9) or (11) to find the derivative of the function on the curve(s).

21. $f(z) = x^2 + y^2 + 2ixy$; x-axis

22. $f(z) = 3x^2y^2 - 6ix^2y^2$; coordinate axes

23. $f(z) = x^3 + 3xy^2 - x + i(y^3 + 3x^2y - y)$; coordinate axes

24. $f(z) = x^2 - x + y + i(y^2 - 5y - x)$; $y = x + 2$

25. In Section 2.1 we defined the complex exponential function $f(z) = e^z$ in the following manner $e^z = e^x\cos y + ie^x\sin y$.

(**a**) Show that $f(z) = e^z$ is an entire function.

(**b**) Show that $f'(z) = f(z)$

26. In Section 3.1 we defined the principal branch of the complex square root in the following manner $z^{1/2} = \sqrt{r}e^{i\theta/2}$, $-\pi < \theta < \pi$.

(**a**) Use (10) to show that $f_1(z) = z^{1/2}$ is analytic in the domain $-\pi < \theta < \pi$.

(**b**) Use (11) to show that $f_1'(z) = \frac{1}{2}z^{-1/2}$ in this domain.

Focus on Concepts

27. Suppose $f(z)$ is analytic. Show that $|f'(z)|^2 = u_x^2 + v_x^2 = u_y^2 + v_y^2$.

28. Suppose $u(x, y)$ and $v(x, y)$ are the real and imaginary parts of an analytic function f. Can $g(z) = v(x, y) + iu(x, y)$ be an analytic function? Discuss and defend your answer with sound mathematics.

29. Suppose $f(z)$ is entire. Use the Cauchy-Riemann equations to show that the function $g(z) = \overline{f(\overline{z})}$ is entire.

30. If $f(z)$ and $\overline{f(z)}$ are both analytic in a domain D, then what can be said about f throughout D?

31. In this problem you are guided through the start of the proof of the proposition:

If f is analytic in a domain D, and $|f(z)| = c$, where c is constant, then f is constant throughout D.

Proof We begin with the hypothesis that $|f(z)| = c$. If $f(z) = u(x, y) + iv(x, y)$ then $|f(z)|^2 = c^2$ is the same as $u^2 + v^2 = c^2$. The partial derivatives of the last expression with respect to x and y are, respectively,

$$2uu_x + 2vv_x = 0 \quad \text{and} \quad 2uu_y + 2vv_y = 0.$$

Complete the proof by using the Cauchy-Riemann equations to replace v_x and v_y in the last pair of equations. Then solve for u_x and u_y and draw a conclusion. Use the Cauchy-Riemann equations again and solve for v_x and v_y.

32. If $f(z) = \overline{z}/z$, then show that $|f(z)|$ is constant in the domain defined by $z \neq 0$. Explain why this does not contradict Theorem 3.3.3(*i*).

33. Suppose $g(x)$ is a twice differentiable real function and f is a complex function defined by $f(z) = g(x) + iy^2$. Determine where $f(z)$ is differentiable and where it is analytic. Where did you use the hypothesis that g is twice differentiable?

34. Find a nonconstant complex function f that is nowhere analytic but is differentiable on the unit circle $|z| = 1$. [*Hint*: Consider using a function f similar to the one in Problem 33.]

35. In this problem you are guided through the start of the proof of the proposition:

If f is analytic in a domain D, and $f'(z) = 0$, then f is constant throughout D.

Proof We begin with the hypothesis that f is analytic in D and hence it is differentiable throughout D. Hence by (9) of this section and the assumption that $f'(z) = 0$ in D, we have $f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x} = 0$. Now complete the proof.

36. Use the Cauchy-Riemann equations to prove that if f is analytic in a domain D, and if $f'(z)$ is a constant, then $f(z)$ must be a complex linear function throughout D.

37. Use the proposition in Problem 35 to show that if f and g are analytic and $f'(z) = g'(z)$, then $f(z) = g(z) + c$, where c is a constant. [*Hint*: Form $h(z) = f(z) - g(z)$.]

38. Suppose $x = r\cos\theta$, $y = r\sin\theta$, and $f(z) = u(x, y) + iv(x, y)$. Show that

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta, \qquad \frac{\partial u}{\partial \theta} = -\frac{\partial u}{\partial x}r\sin\theta + \frac{\partial u}{\partial y}r\cos\theta \tag{12}$$

and

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta, \qquad \frac{\partial v}{\partial \theta} = -\frac{\partial v}{\partial x}r\sin\theta + \frac{\partial v}{\partial y}r\cos\theta. \tag{13}$$